Ellipses

Section 1: Horizontal and Vertical Standard Forms

1 Find the center, vertices, and co-vertices of $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

a. Center: $(0, 0)$, Vertices: $(\pm 5, 0)$, Co-vertices: $(0, \pm 3)$

b. Center: $(0, 0)$, Vertices: $(0, \pm 5)$, Co-vertices: $(\pm 3, 0)$

c. Center: $(5, 3)$, Vertices: $(\pm 5, 0)$

d. Center: $(0, 0)$, Vertices: $(\pm 25, 0)$

2 Find the standard equation of the ellipse with center $(0, 0)$, vertices $(0, \pm 6)$, and co-vertices $(\pm 4, 0)$.

a. $\frac{x^2}{16} + \frac{y^2}{36} = 1$

b. $\frac{x^2}{36} + \frac{y^2}{16} = 1$

c. $\frac{x^2}{4} + \frac{y^2}{6} = 1$

d. $\frac{x^2}{20} + \frac{y^2}{36} = 1$

3 Find the length of the major and minor axes for $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{4} = 1$.

a. Major axis $= 8$, Minor axis $= 4$

b. Major axis $= 4$, Minor axis $= 2$

c. Major axis $= 16$, Minor axis $= 4$

d. Major axis $= 8$, Minor axis $= 2$

Section 2: Foci and the Fundamental Relationship

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4 What is the fundamental relationship between semi-major axis a, semi-minor axis b, and focal distance c in an ellipse?

a. $c^2 = a^2 - b^2$

b. $c^2 = a^2 + b^2$

c. $a^2 = b^2 - c^2$

d. $c = a - b$

5 Find the coordinates of the foci of $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

a. $(\pm 4, 0)$

b. $(0, \pm 4)$

c. $(\pm\sqrt{34}, 0)$

d. $(\pm 16, 0)$

6 Find the foci of $\frac{x^2}{16} + \frac{y^2}{25} = 1$.

a. $(0, \pm 3)$

b. $(\pm 3, 0)$

c. $(0, \pm\sqrt{41})$

d. $(\pm 9, 0)$

Section 3: Eccentricity

7 How is the eccentricity e of an ellipse defined, and what is its range?

a. $e = \frac{c}{a}$, where $0 \le e < 1$

b. $e = \frac{a}{c}$, where $e > 1$

c. $e = \frac{b}{a}$, where $e < 0$

d. $e = c^2/a^2$

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8 Find the eccentricity of the ellipse $\frac{x^2}{100} + \frac{y^2}{64} = 1$.

a. 0.6

b. 0.8

c. 0.36

d. 0.64

9 What happens to the shape of an ellipse as its eccentricity approaches 0?

a. It becomes nearly circular

b. It becomes very elongated / needle-like

c. It approaches a parabola

d. It opens outward

Section 4: Completing the Square and Whispering Galleries

10 Convert $4x^2 + 9y^2 - 16x + 54y + 61 = 0$ to standard form.

a. $\frac{(x-2)^2}{9} + \frac{(y+3)^2}{4} = 1$

b. $\frac{(x-2)^2}{4} + \frac{(y+3)^2}{9} = 1$

c. $\frac{(x+2)^2}{9} + \frac{(y-3)^2}{4} = 1$

d. $(x-2)^2 + (y+3)^2 = 36$

11 A whispering gallery is an elliptical room 100 ft long and 60 ft wide. How far from the center are the two whispering spots (foci)?

a. 40 ft

b. 50 ft

c. 30 ft

d. 20 ft

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12 Why can a whisper at one focus of an elliptical gallery be heard clearly at the other focus?

a. Sound waves reflecting off the elliptical wall all converge simultaneously at the other focus

b. Sound waves travel faster along ellipses

c. The walls absorb echo

d. The ceiling acts as a resonator

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